populace dynamics: an open, scored longitudinal layer for policy microsimulation

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Abstract

Dynamic microsimulation — the longitudinal aging of a person-level population through earnings, family structure, disability, mortality, and program participation — underpins retirement and social-insurance policy analysis, yet every major United States model of this kind is closed: internal to government, tied to restricted administrative records, or accessible only through institutional relationships. This design paper specifies an open alternative built as an extension of populace, PolicyEngine's open-source microdata stack, whose cross-sectional layer is the certified default United States microdata in PolicyEngine after outperforming its predecessor on held-out administrative targets. The design contributes four elements: a trajectory-weighted kernel in which multi-period calibration cannot silently destroy panel structure; a Dynamics operator that treats state transitions as conditional models, mixing deterministic demographic hazards with machine-learned earnings processes; an explicit domains-of-validity framework that refuses point forecasts where parameter uncertainty dominates — including the 75-year actuarial balance — and instead publishes sensitivity surfaces; and a scoring protocol under which every claim either resolves against administrative publications, is backtested with leakage control, or is computed exactly from statute, with contributions merging only when they improve held-out scores. United States Social Security is the first validation domain; the layer itself is country-agnostic.

1 Introduction

Public microsimulation infrastructure is unevenly distributed. On the tax side, analysts can run open, calibrated models directly: Tax-Calculator (Policy Simulation Library 2026), PolicyEngine (PolicyEngine 2026), and FiscalSim-US (Center for Growth and Opportunity at Utah State University 2026) are openly callable on public data, alongside source-available models with restricted inputs (The Budget Lab at Yale 2026) and proprietary models used for outside-facing analysis (Tax Policy Center 2025; Institute on Taxation and Economic Policy 2025; Tax Foundation 2025; Penn Wharton Budget Model 2025). On the dynamic side — the longitudinal modeling that retirement and social-insurance policy requires — the benchmark models remain closed. DYNASIM (Favreault et al. 2015; Urban Institute 2024), MINT (Social Security Administration 2024), CBOLT (Congressional Budget Office 2018, 2024), and Morningstar's retirement-outcomes model (Look and VanDerhei 2024) are real and important, but outside users reach them only through institutional relationships. The nearest open analogue, the Cato Social Security model (Chanwong 2026), demonstrates feasibility while remaining deliberately narrow in population, scope, and validation.

This paper specifies the design of an open longitudinal layer — *populace dynamics* — as an extension of populace, PolicyEngine's open-source microdata stack. It is a design paper: the cross-sectional foundation is shipped and validated; the longitudinal layer is specified here and built in the open, gated by the scoring protocol of Section 5.

The design makes four contributions:

1. **A trajectory-weighted kernel with explicit alignment.** One weight per trajectory, with multi-period calibration expressed as stacked constraint rows over the same weight vector, so that hitting cross-sectional totals in multiple periods cannot silently destroy panel structure — combined with event-selection alignment for period-by-period control, since weights alone cannot separate a correct life course from a correctly timed one (Section 4).
2. **Transitions as conditional models.** A single operator interface covers deterministic demographic hazards and machine-learned earnings processes, so candidate architectures compete under one evaluation standard (Section 4).
3. **Domains of validity as shipped metadata.** The model states which questions it will not answer with false precision — led by the 75-year actuarial balance, where input uncertainty dominates model fidelity — and publishes long-horizon results as sensitivity surfaces rather than point forecasts (Section 3).
4. **A scoring protocol in place of fidelity-only validation.** Claims resolve against administrative publications on an annual calendar, backtest against realized history with leakage control, or compute exactly from statute; contributions merge only when they improve held-out scores (Section 5).

United States Social Security is the first validation domain, chosen because it is the hardest available proving ground: benefit adequacy and reform incidence depend jointly on lifetime earnings, marriage and survivorship, disability, differential mortality, and claiming. The layer itself is country-agnostic and extends to other pension and benefit systems as PolicyEngine's country coverage grows.

2 Terminology and scope

"Dynamic" is used here in the microsimulation field's standard sense, following Orcutt et al. (1961) and the tradition carried by DYNASIM, MINT, CBOLT, and SimPaths: a longitudinal model that ages a person-level population through time (Li and O'Donoghue 2013). It does not mean "dynamic scoring" — the tax-policy usage denoting macroeconomic feedback in revenue estimation — and the design is not an overlapping-generations general-equilibrium model of the Auerbach–Kotlikoff kind, such as the Penn Wharton Budget Model operates (Penn Wharton Budget Model 2025). The model claims neither macroeconomic feedback nor equilibrium closure. Behavioral responses enter as labeled scenario inputs with documented ranges, not as point estimates carrying model authority.

3 Domains of validity

All models are wrong; a model earns its keep only where it improves predictions. The design therefore begins by bounding its own claims.

3.1 Parameter uncertainty dominates the long horizon

The 75-year actuarial balance of a pension system is a function of a small set of exogenous assumptions — fertility, mortality improvement, net immigration, real wage growth, interest rates — whose uncertainty dominates the microsimulation machinery that processes them. The public record makes this concrete for United States Social Security. The Trustees' own low- and high-cost scenarios bracket a range of 75-year balances wider than the intermediate deficit itself (Board of Trustees, Federal Old-Age and Survivors Insurance and Federal Disability Insurance Trust Funds 2025). The Congressional Budget Office's long-term projections differ materially from the Trustees' — chiefly for assumption reasons, though method differences such as CBO's micro-founded projection of the taxable share of earnings also do real work in the divergence (Congressional Budget Office 2024). Assumption variance dominates the headline; model structure matters for distribution and the near term, which is the division of labor the output tiers encode. Successive Technical Panels convened by the Social Security Advisory Board have recommended revising fertility and mortality-improvement assumptions as realized values ran persistently outside the projected path (Technical Panel on Assumptions and Methods 2023). Two expert institutions with administrative data and decades of refinement disagree with each other and have both missed realized demographic trends; a better microsimulation does not repair that, because the variance lives in the inputs.

3.2 Three output tiers

The model's outputs are organized into tiers, each attached to the strongest claim it can support.

Tier 1: distributional analysis under fixed assumptions. Reform analysis is a difference — outcome under reform minus outcome under baseline, holding the population and assumption path fixed — and much of the unforecastable demographic uncertainty is common to both arms and cancels in the difference. The required ingredients — a calibrated joint distribution of lifetime earnings, family structure, and differential mortality, plus an exact rules engine — are the components that can be validated directly. The slice of a reform delta that does not cancel is labeled as the least-validated slice rather than buried: reform-induced claiming responses, and interactions between the reform and uncertain dynamics — the mortality gradient enters many deltas as a covariance with benefit position, not a level, so it does not difference out. Deltas evaluated past trust-fund depletion also require an explicit scheduled-versus-payable baseline convention, which the model states with every such output. For claiming specifically, scenario ranges anchor to the quasi-experimental record on retirement-age responses (Mastrobuoni 2009; Behaghel and Blau 2012), published as a scenario library rather than embedded as point estimates.

Tier 2: near-term components that resolve. Over roughly a ten-year horizon, outputs are dominated by mechanics rather than demographic extrapolation: beneficiary counts by type, average benefits, covered earnings and taxable payroll, claiming-age distributions, disability incidence. These resolve against administrative publications on an annual cycle and are scored (Section 5).

Tier 3: the long horizon as a sensitivity surface. Long-horizon outputs are published as surfaces over documented assumption ranges — how the balance, cohort replacement rates, or distributional outcomes move as fertility, mortality improvement, and immigration vary — never as point forecasts. Incumbent practice buries assumption-dependence behind a point estimate and an appendix; an open model can invert that and make the sensitivity the interface.

Every API response carries its tier, assumption path, and calibration history as metadata, so a downstream consumer — human or machine — weights the output by demonstrated reliability rather than by the producer's reputation.

4 Architecture

4.1 The cross-sectional foundation

populace builds a calibrated synthetic population entirely from primary-source government data — the Current Population Survey ASEC, the IRS Public Use File, the Survey of Consumer Finances, SIPP, CPS outgoing-rotation groups, MEPS, and the ACS — synthesizing missing variables with weight-aware conditional models and calibrating to administrative aggregates treated as uncertainty-weighted facts. In June 2026 it replaced PolicyEngine's enhanced CPS as the certified default United States microdata in PolicyEngine, after a matched, symmetric-refit comparison on 41,314 households with a 739-target holdout (Table 1).

Table 1: Certification comparison from the populace release manifest. The per-target asymmetry is reported deliberately: the enhanced CPS wins more individual targets narrowly while its largest misses are far larger, and publishing the number that cuts against the headline is the discipline the scoring protocol requires.

Metric (lower is better)	populace	enhanced CPS
Holdout loss (739 held-out targets)	0.038	0.317
Training loss	0.190	1.089
Full loss	0.228	1.405
Per-target wins	1,040	2,613 (51 ties)

4.2 Trajectory weights and population accounting

The longitudinal extension follows two kernel rules set in populace's charter. First, **one weight per trajectory**: multi-period calibration targets stack as (target, period) constraint rows over a single trajectory-level weight vector, so that calibrating cross-sections independently — the standard way panels die — is a kernel-level error rather than a modeling temptation. Second, **population is not closed**: trajectories carry entry and exit markers (birth, death, immigration, emigration), and a trajectory's weight contributes to a period only while the person is present. Household and couple links are period-scoped, so family recomposition preserves per-period accounting identities; couples carry a shared unit weight derived from their trajectory weights so that spousal and survivor benefits have a well-defined representation.

Weights alone are one layer of alignment, not the whole answer: a weight cannot distinguish a correct life course from a correctly timed one, and reweighting trajectories to hit future cross-sectional cells risks selecting on entire correlated life courses. Period-by-period control therefore also operates through event selection — ranking individual transition probabilities and selecting the number of events an external control demands, the mechanism CBOLT and DYNASIM use — with trajectory weights reserved for base-year representation and slow-moving composition (Li and O'Donoghue 2013; Dekkers and Cumpston 2012). The stacked constraint system is solved as uncertainty-weighted penalized least squares against target standard errors, so infeasible combinations resolve by SE-weighted compromise rather than silent failure, and cohorts born after the base year are pinned by projected demographic controls rather than left free.

4.3 The Dynamics operator

Dynamics is an operator from a population and a transition specification to a population with extended periods. A transition is a conditional distribution — the probability of next-period state given current state and covariates — which is the same interface populace's synthesis models already implement. The shipped baseline for earnings is a regime-gated, sequentially chained, weighted quantile-regression-forest imputer (Meinshausen 2006) whose zero-inflation gate doubles as a nonemployment model; richer architectures (zero-inflated neural distribution models, normalizing flows) are candidates that must beat the baseline on held-out longitudinal moments to merge.

The design is deliberately hybrid. Where the evidence base is tabular — mortality from official life tables with published income gradients, fertility from vital statistics, marriage and divorce from ACS- and CPS-based rates (federal collection of detailed marriage and divorce statistics ended in the 1990s), disability incidence from program statistics — transitions are deterministic hazards, auditable row by row. Marriage requires a matching model, not only a hazard: spousal and survivor benefits depend on assortative matching over lifetime earnings, which the design treats as a first-class estimation target rather than an afterthought. Machine learning is reserved for processes with rich conditional structure, led by earnings dynamics — where the process is not stationary: volatility and mobility vary by cohort and period in administrative data (Sabelhaus and Song 2010; Kopczuk et al. 2010), so transition models carry cohort and period conditioning rather than pooling across decades. Chained one-period models also understate long-spell persistence unless spell structure is modeled explicitly, and backcasting is a distinct conditional object from forward simulation rather than the same operator reversed; both are treated as evaluation targets in their own right, disciplined by held-out panel moments such as higher-order earnings-change distributions (Guvenen et al. 2021).

One measurement caveat is stated rather than hidden: the most demanding earnings-dynamics moments come from administrative records that public panels understate, and survey- and administrative-based estimates disagree on volatility levels and trends. Where a published administrative moment cannot be recomputed on held-out public data, matching it is calibration, not validation, and the scorecard labels it as such.

4.4 Rules and delivery

Statute is the deterministic slice of any policy forecast. Core retirement benefit formulas — average indexed monthly earnings, primary insurance amounts, actuarial adjustments — and benefit taxation compute exactly today through PolicyEngine's rules engine via populace's rules-adapter protocol, vectorized over person-periods. Auxiliary, spousal, and survivor benefit formulas are explicit build items in the validation program: the current engine carries them as calibrated aggregates rather than person-level formulas, and the scorecard treats "computes exactly" as a per-rule status to be earned, not a blanket claim.

Data governance is a design requirement, not an afterthought. Estimating transition models on restricted-use panels and publishing the estimated parameters is settled practice — open models such as OG-USA, and DYNASIM itself, estimate on the PSID and publish what they learn. Releasing a synthetic *microdata* artifact informed by such panels is a stricter problem, because donor-based samplers can emit observed training values. The design answers it structurally: released records originate from populace's public-use cross-section, restricted panels train processes rather than donate records, samplers are smoothed or noised so no verbatim donor values ship, and every release passes nearest-neighbor disclosure checks alongside its accuracy scorecard — with data producers engaged directly where their terms of use warrant it. Uncertainty budgets therefore attach only to the genuinely uncertain components. The deliverable is a versioned artifact — a longitudinal population release with a manifest and scorecard, certified through the same path as the cross-sectional release — exposed through a Python library, a REST API, and a Model Context Protocol server so that AI agents can run baseline distributions and reform analyses with validity metadata attached.

5 Scoring and resolution

Most microsimulation validation asks whether the model matches published aggregates. That standard is necessary but weak: a model can be tuned to reproduce the tables it was built from. This project's standard is that a claim is validated when it improves prediction of something that subsequently resolves. Five scoring surfaces implement it.

1. **Annually resolving components.** Beneficiary counts by type, average and aggregate benefits, covered earnings and taxable payroll, cost-of-living adjustments, disability incidence, and claiming-age distributions resolve against administrative publications each year. Every published forecast cell carries a resolution rule naming the exact table and vintage that settles it.
2. **Forecasting the forecasters.** Official projections revise constantly; predicting the next revision of headline quantities resolves in months rather than decades and is decision-relevant to anyone who acts on the official number. This is partly forecasting an assumptions process — panels advise, committees adopt with a lag — so each cell pre-specifies the naive baseline it must beat (an assumption random walk plus mechanical data update).

3. **Retrodiction with leakage control.** The model is built from data vintages available at a historical date and scored against realized outcomes. populace's versioned data registry pins vintages going forward; pre-registry history is a reconstruction problem — survey redesigns, revised administrative tables, re-released panels — so pre-registry backtests are graded as pseudo-vintage, with a published log of deviations from true vintage, and no registry pins specification leakage: a "2005-vintage" model built today knows 2008 happened, and the protocol says so. Retrodictive calibration under the historical regime does not guarantee calibration under a new one, so backtests complement rather than substitute for live resolution.
4. **Statutory resolution.** Where an output is fixed by law, the rules engine computes it exactly, and enacted policy settles the corresponding conditional cells immediately.
5. **Held-out panel moments.** The population layer is scored against moments it was not fit to: earnings-mobility matrices, autocorrelation and higher-order moments of earnings changes, cohort age-earnings profiles, and family-transition rates on held-out panel records.

Two governance rules complete the protocol. **Merge on score**: a contribution — a mortality module, a claiming model, an earnings architecture, from any contributor — merges if and only if it improves the population's score on held-out facts, the rule populace already applies to its cross-sectional layer. **Publication discipline**: misses publish with the same prominence as hits; superseded methods keep their historical scorecards; and stage gates in the development roadmap are pre-specified score thresholds, not narrative judgments.

6 First validation domain: U.S. Social Security

Social Security is the first domain because it is the hardest available test of the machinery: eligibility and benefits depend on the highest thirty-five years of indexed earnings, marital and survivorship histories, disability pathways, differential mortality, and claiming timing — jointly. A layer that scores well here earns reuse in adjacent domains — Supplemental Security Income interactions, long-term care, and retirement adequacy, the last of which is *harder*, not easier: wealth projection was the weak link of even administrative-data models (Favreault and Smith 2016), and a wealth and pension roadmap is future work, not an assumed extension — and in other countries' pension systems.

The domain also sharpens the honesty argument. The canonical output of existing Social Security models — the 75-year balance and depletion date — is precisely the quantity Section 3 declines to forecast. What the open layer offers instead is the combination the field lacks: tier-1 distributional incidence of reforms with reproducible assumptions; tier-2 near-term components with a public resolution record; and tier-3 sensitivity surfaces that make the assumption-dependence of long-horizon claims the interface rather than the appendix. No existing model, closed or open, publishes that combination. The closed benchmarks do publish validation and cohort tables; what outsiders cannot do is rerun the pipeline, vary its assumptions, or audit the intermediate states behind the published numbers.

7 Related work

Dynamic microsimulation originates with Orcutt et al. (1961); Li and O'Donoghue (2013) survey the field's alignment and calibration practice. DYNASIM (Favreault et al. 2015; Urban Institute 2024), MINT (Social Security Administration 2024), and CBOLT (Congressional Budget Office 2018) define the closed state of the art in the United States, and this design borrows their central structural lesson — an annual state engine with family links and external alignment — while departing on openness, trajectory-level weighting, and scoring; Dekkers and Cumpston (2012) treat the weighting question directly. The Cato model (Chanwong 2026) is the nearest open system and demonstrates demand for one. On the measurement side, administrative-data studies of earnings dynamics discipline the earnings process: lifecycle moments (Guvenen et al. 2021), long-run mobility (Kopczuk et al. 2010), non-stationary volatility (Sabelhaus and Song 2010), and the attenuating link between current and lifetime earnings (Haider and Solon 2006) — with nonlinear panel frameworks (Arellano et al. 2017) the academic cousin of the quantile machinery used here. The quasi-experimental record on claiming responses (Mastrobuoni 2009; Behaghel and Blau 2012) anchors the behavioral scenario library, and the assumption-failure record compiled by successive Technical Panels (Technical Panel on Assumptions and Methods 2023) motivates the domains-of-validity framework.

8 Status and roadmap

The cross-sectional foundation is in production. The longitudinal kernel rules are specified in populace's charter, and this paper's companion repository holds the full design book: operational chapters on earnings-history construction, family and auxiliary benefits, disability and claiming, mortality and projection drift, calibration targets, and the Social Security validation program. Development proceeds through stage gates defined as score thresholds — earnings-history credibility first, family and benefit outputs second, forward projection third, productization last — with every gate's evidence published whether it passes or fails.

The layer is open source under the MIT license. Corrections are invited, particularly to the benchmark characterizations, and contributions merge on the same standard as the authors': improve the held-out score.

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